

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		all have 7 electrons in outer shell so need to gain 1 electron (1) reactivity decreases down the group because outer shell becomes further from nucleus / has more shielding (1) therefore becomes more difficult to attract an electron (1)	3			3		
		(ii)		$4 \text{ HF} + \text{SiO}_2 \rightarrow \text{SiF}_4 + 2 \text{ H}_2\text{O}$		1		1	1	
		(iii)		$\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow 2\text{HF} + \text{CaSO}_4$ reactants (1) products (1) balancing (1) award only when reactants are products are correct		3		3	1	
	(b)			mass chlorine = 21.25g (1) $\frac{21.25}{35.5}$ and $\frac{5.45}{27}$ (1) 0.599 : 0.202 ratio 1 : 3 therefore formula AlCl_3 (1) must show working		3		3	3	
				Question 7 total	3	7	0	10	5	0